Homework 4

GENERAL ASSIGNMENT INSTRUCTIONS regarding the use of AI tools

AI tools like ChatGPT can be a powerful resource for generating ideas and refining your work. However, it is essential to know how to use these tools effectively and critically. Make sure to verify the relevance and accuracy of the answers you receive and be mindful of how AI-generated content aligns with the requirements of the assignment. Your own judgment and understanding are key to producing high-quality work.

Along with your responses, please append a list of any requests made to AI tools, such as ChatGPT, if used during the process of creating your answer.

Problem 1 (Quality of life): The data for this problem come from a study of quality of life in women treated for metastatic breast cancer. Women were randomly assigned to a control group who received "usual care" and a treatment group who received weekly sessions of group therapy and self-hypnosis. Follow up continued for 122 months (just longer than 10 years). The study was set up to look at quality of life, but after 10 years, it was noticed that women in the treatment group appeared to live longer than those in the control group. This analysis follows up on longevity. Most women died during the follow up period, but 3 women were still alive.

The data are in cancer.csv and the four variables are:

- *Survival*: survival time, in months
- *Group*: randomly assigned treatment (Control or Therapy)
- *Censor*: 0 for a woman who died, 1 for a woman who was alive at the 122 month follow-up
- *SurvivalC*: survival time, in months, with those three living women changed to 123. This change was made so that the censored individuals (survival > 122 months) have a value larger than 122.

a) Draw a side-by-side boxplot of the survival times in the two groups of women. Your answer for this question is the plot.

b) Estimate and report the **median** survival time for each of the two groups of women.

c) Use an appropriate method to test the null hypothesis that the treatment has no effect on the median survival time. Report your p-value and an appropriate one-sentence conclusion from this test.

d) Give two reasons why it is more appropriate to report medians, not means, for these data. *For at least one of the reasons, remember that most medical studies make recommendations for individuals.*

Problem 2 (Cross/self-pollination) The dataset provided in the file darwin.csv represents a small subset of the extensive studies conducted by Charles Darwin on cross-fertilized and self-fertilized plants. Darwin documented the entirety of his experiments in his 1878 publication. The data in this subset pertain specifically to corn plants. In his experiments, Darwin manually pollinated plants to produce both cross-fertilized seedlings and self-fertilized seedlings. For each experimental pair, one cross-fertilized seedling and one self-fertilized seedling were planted side by side in individual pots and placed in a greenhouse. Darwin stated that the environmental conditions experienced by each pair of plants were virtually identical, ensuring that any observed differences in growth could be attributed primarily to the method of fertilization rather than to environmental variation.

a) Draw a dotplot of the differences (as cross - self). Your answer is the plot.

b) What, if anything, in the dotplot suggests that a *paired t*-test is not appropriate? Briefly explain your answer.

c) Use an appropriate non-parametric method to test whether there is a difference between self- and cross-fertilized seedlings. Report your p-value and an appropriate one-sentence conclusion, in the context of this study.

Problem 3 (Hormone Therapy):

The Women's Health Initiative conducted a randomized experiment to see if hormone therapy was helpful for post-menopausal women. The women were randomly assigned to receive the estrogen plus progestin hormone therapy or a placebo. After 5 years, the number of women who developed cancer in each group was determined. The data can be found in the WHI.csv file in Canvas.

a) Obtain a contingency table of the two variables. Estimate and report the proportion of women who developed cancer in each group.
b) Estimate the standard error of the proportion of women in the treatment group who developed cancer.
c) Estimate the odds that a woman in the treatment group will develop cancer.
d) Conduct a hypothesis test to determine if the proportion of women with cancer is different between the two groups.
e) Calculate a 90% confidence interval for the difference in the proportion of women with cancer between the hormone therapy group and the placebo group. Interpret this confidence interval.
f) Calculate a 90% confidence interval for the odds ratio of developing cancer when taking hormone therapy and interpret this confidence interval.
g) Use a Chi-square test to test the hypothesis that the women in the control and treatment group are equally likely to get cancer. Do not use the continuity correction. Report the Chi-square statistic, *p*-value, and a one-sentence meaningful conclusion in the context of this study.

Problem 4 (Political polls): In 1990, a random sample of 1,600 adults in the United States were asked whether they approved of the job George H.W. Bush was doing as president. One month later, the same people were asked the same question. The data are given in the file Bushapproval.csv.

a) Calculate the contingency table for the two approval ratings.
b) Use the contingency table to calculate an estimate of the proportion of adults in the U.S. who approved of the job President Bush was doing for each time period.
c) Explain why the two estimates you calculated in part (b) are not independent from each other.
d) Use McNemar's test to determine if the proportion of adults in the U.S. who approved of the job President Bush was doing is different between the two time periods.
e) Calculate a 95% confidence interval for the difference in the proportion of adults in the U.S. who approved of the job President Bush was doing between the two time periods. Provide a meaningful interpretation of this confidence interval within the context of this study.

General homework notes:

1. Be sure to check which way around R is computing the difference.
2. For questions requesting numeric answers, copy the appropriate numbers from computer output and report those in your homework submission.
3. Do NOT hand in copies of computer output. You should keep a copy of your code so you can compare with our code, in case you get different results.
4. If you use Rmarkdown (or similar), please do not include the Rmarkdown output (i.e., code + unfiltered results). See advice point 2).